Beyond Topics: Subcommunities in Substack's Recommendation Network

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ABSTRACT

Substack is a leading platform for independent publishing, where authors actively recommend other newsletters to their readership. Unlike algorithmic suggestions, these links represent deliberate editorial endorsements. While one might expect these endorsements to reflect shared subject matter, they may also signal shared views. This study investigates whether recommendation communities merely mirror newsletter topics or also capture affinities that topic labels miss. To address this, the recommendations were modeled as a network constructed via a breadth-first snowball crawl from 15 seed newsletters across five domains, yielding 15,186 nodes and 103,425 edges. Newsletter topics were inferred from post texts using TF–IDF and k-means clustering, while network communities were detected using the Louvain algorithm and validated for stability with Leiden. The analysis reveals a small-world, heavy-tailed network topology. Overall, recommendations strongly align with their topics: in domains like finance and food, a single community encompasses approximately 90% of newsletters on that subject. However, where positions conflict, the topic alone is insufficient to explain network structure. Political newsletters fracture into three distinct communities (liberal mainstream, heterodox-centrist, and anti-establishment) that cut across finer sub-topics, and health newsletters critical of COVID-19 vaccines cluster with anti-establishment political commentary rather than with public-health publications.
[1]

KEYWORDS

Social Network Analysis, Substack, recommendation networks, community detection, homophily

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1 INTRODUCTION

In recent years, many journalists, writers and experts have supplemented their work at traditional outlets with independent publishing, often through newsletters delivered directly to subscribers' inboxes and supported by paid subscriptions. This shift means that part of public debate now takes place not in newspapers, which bring together many voices under one editorial line, but in a fragmented landscape of individual authors, each addressing their own audience. How these authors relate to one another is therefore an open question: whether they form communities of shared interests, or clusters of like-minded voices that rarely cross. On Substack, this question can be observed directly, because authors publicly point their readers to other publications they value.

These recommendations are expected to follow subject matter, in the classic pattern of homophily [4]. Subject, however, may not be the whole story: Adamic and Glance found that political blogs linked mostly to blogs with a similar orientation [1]. This study asks *whether the communities formed by recommendations simply mirror topics, or whether in some domains they separate newsletters that write about the same subject.*

2 DATA COLLECTION

To study whether recommendations follow topics or positions, it was necessary to build a dataset that reflects both the recommendation choices of authors and the content of their newsletters.

[1] **Project Repositories**
Data Collection: https://github.com/sara-rss/substack-recommendation-network/tree/main/data_collection
Network Analysis: https://github.com/sara-rss/substack-recommendation-network/tree/main/network_analysis
Open Problem: https://github.com/sara-rss/substack-recommendation-network/tree/main/open_problem

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Selected Data Sources

The data were collected from Substack, which makes publicly accessible both the recommendations of each newsletter and its recent posts. Recommendation lists were retrieved with the unofficial Python library `substack-api` [3], and the most recent posts of each newsletter through the public `/api/v1/archive` endpoint. Substack does not, however, expose the category of newsletters run by other authors, so the topic of each newsletter had to be inferred from its posts (see Node Attributes below). From these data, a network was built in which nodes are newsletters and a directed edge $A \to B$ indicates that A recommends B.

The crawl began with 15 manually selected *seed* newsletters, three for each of five broad areas: literature, politics, technology, finance and health. Within each area, well-known and active newsletters were chosen, so as to broaden thematic coverage and reduce the risk of concentrating the crawl in a single niche. Each seed was checked for active recommendations.

Crawling Methodology and Assumptions. The network was explored through snowball sampling, the only option in the absence of a complete list of Substack publications. The crawl proceeded breadth-first: for each newsletter it visited, the crawler downloaded the full list of publications it recommends and added the new ones to a queue, to be visited in turn.

The same newsletter may appear under slightly different URLs, for instance, with or without `www`, or with `http` rather than `https`. Each URL was therefore normalized before the network was constructed, ensuring that every newsletter corresponded to a single node.

To avoid overloading the platform, the crawler waited one second between requests, and its progress was saved every 100 newsletters so that the crawl could be resumed after any interruption. Despite these precautions, the requests for 59 newsletters failed because of connection drops; they were repeated in a second pass, and all of them were recovered.

The network analyzed reflects three methodological choices:

- **Induced subgraph.** Only expanded newsletters, whose recommendation lists were actually downloaded, are kept; recommendations pointing to the 38,525 newsletters discovered but never visited are discarded.
- **Static snapshot.** Recommendations were collected once, in August 2026, and the network describes the platform at that time.
- **Undirected graph.** The analyses use the undirected, unweighted version of the network. Since 41% of recommendations are reciprocated, this discards the direction of the remaining 59%.

Node Attributes

The topic of each newsletter was inferred from the content of its posts. Titles and subtitles of the ten most recent posts were collected for every newsletter; 13,816 of the 15,186 newsletters (91%) returned enough text to be analyzed. Topics were then identified in four steps:

(1) **Language detection.** The language of each newsletter was detected with `langdetect` [2, 5]. The 387 newsletters not in English (2.5% of the network) were kept with language as a separate attribute, since a text model built for English would group them by language rather than by topic.
(2) **Clustering.** After removing standard stopwords, numbers, dates, and generic newsletter terms, English texts were vectorized with TF–IDF [7] and clustered via k-means. Neither the elbow method nor the silhouette score points to an optimal number of clusters: across the values of k tested between 5 and 30, the inertia decreases steadily by only 1.5%, without a clear elbow, and the silhouette never exceeds 0.004, as is typical of sparse, high-dimensional text vectors. The number of clusters was set to $k = 16$, the smallest value where all five seed areas emerge as distinct clusters and health separates into its own topic.
(3) **Recovery from post bodies.** Three clusters are weakly informative: generic titles, video and podcast episodes, and personal diaries. For their 5,097 members, the beginning of the post bodies was also downloaded; the 5,082 with enough body text were reassigned to the nearest existing centroid, without re-running k-means. This moved 1,246 newsletters to a more specific topic.
(4) **Keyword dictionary.** A dictionary of unambiguous keywords was applied to the newsletters remaining in these three clusters, only when at least 1% of the words were topical and the first topic outweighed the second by a factor of 1.5. This moved 445 more.

The resulting clusters were named by inspecting their most characteristic terms and grouped into eight macro-topics (Table 1). Overall, 11,886 newsletters (78.3%) have a topic label. This figure includes the 1,863 newsletters left in the *personal* cluster, which is kept as a topic within lifestyle and culture although it is only weakly informative; generic and video-only newsletters are instead treated as having no topic. Since the area of each seed was chosen in advance, the 15 seeds offer a small consistency check: the classification, which ignores this information, assigns 12 of them to their area. Among the errors, Glenn Greenwald's newsletter publishes mostly video episodes whose titles do not reveal their subject, a limit of title-based topics discussed in Section 5.

Table 1: Macro-topics inferred from post texts

Macro-topic	Topics	%
Lifestyle & culture	style and home, personal	19.4
Arts & letters	writing, music	11.5
Technology	AI, AI and coding	11.4
Politics	US, UK, geopolitics	10.9
Finance	equity, macro	10.2
Business	business	7.4
Science & health	health	4.7
Food	food	2.8
No topic	generic, non-English, video	12.7
No text		9.0

3 NETWORK CHARACTERIZATION

The final network comprises 15,186 newsletters and 103,425 recommendations. As explained in Section 2, all measures are computed on the undirected version of the graph, which contains 82,062 edges.

Degree Distribution

Connectivity is highly unequal. The degree ranges from 1 to 380, with a mean of 10.81 but a median of only 6: 15% of newsletters have a single connection, while a few hubs reach several hundred. At most 50 outgoing recommendations per newsletter are available, so high degrees are driven by recommendations received: for the three largest hubs, over 90% of connections are incoming. Heather Cox Richardson's *Letters from an American*, a political newsletter, has the highest degree and is recommended by 377 other newsletters. Other prominent hubs focus on venture capital, start-ups, artificial intelligence, and finance.

The degree distribution was examined through the three panels of Figure 1:

- in the log-log plot (left), $P(k)$ bends downwards as the degree increases;
- the cumulative distribution (center) shows that about two thirds of newsletters have fewer than ten connections;
- the tail was fitted with the `powerlaw` package. The best power law ($k_{\min} = 19$, $\alpha = 2.82$, with $P(k) \propto k^{-\alpha}$) is outperformed by both a lognormal and a truncated power law ($p < 10^{-6}$), but outperforms an exponential ($p = 0.036$). In the complementary cumulative distribution (right), the power law departs from the data around $k = 40$, while the lognormal follows them up to the largest hubs.

The distribution is therefore heavy-tailed but not scale-free in the strict sense. The network is also mildly disassortative ($r = -0.081$), a pattern common in information networks [6] that may partly reflect the asymmetry of recommendations.

Components and Paths

The network forms a single connected component, a result that reflects the snowball sampling, which only reaches newsletters linked to the seeds. Shortest paths were computed between all pairs of nodes. The average shortest path is 4.34 and the diameter of the network is 9, but long paths are rare: 41% of pairs are at distance 4, and 91% are at most five steps apart. Thus most newsletters can be connected through a short chain of recommendations.

Density and Clustering

The network density is 0.00071, making the network very sparse: on average, each newsletter is connected to fewer than 0.1% of the others. The average clustering coefficient is 0.151, modest in absolute terms, and it declines steadily with degree, from 0.25 among newsletters with two to five connections to 0.03 among those with more than one hundred. Thus, lower-degree newsletters tend to have more tightly knit neighborhoods, whereas the neighbors of hubs are rarely connected to one another.

The network contains 57,649 triangles. Its global transitivity (0.064) is lower than the average clustering, a difference that may reflect open triads around hubs: several newsletters are connected to the same hub without being connected to one another, so hubs contribute many connected triples, most of which remain open.

Centrality Analysis

Three centrality measures were computed: *degree*, *betweenness* and *closeness* (Table 2). To compare them, Spearman's rank correlation ρ was used, which measures how similar the rankings produced by two measures are.

Betweenness is strongly correlated with degree ($\rho = 0.913$), while closeness departs more from it ($\rho = 0.827$): next to the largest hubs, it brings out four newsletters that are not among the most connected: Paul Krugman's newsletter, *The Free Press*, *Noahpinion* and *The Honest Broker*, all offering commentary on politics, economics or culture rather than on a single niche.

A central position, however, can have two different meanings. A newsletter may be central because it links different parts of the network, or simply because it sits in the middle of one large community. To tell the two cases apart, the participation coefficient $P = 1 - \sum_c (k_c/k)^2$ was computed, where k_c is the number of neighbors in community c: P is 0 when all the neighbors of a newsletter belong to one community, and approaches 1 when they are spread across many. Its median is 0.30 for newsletters with 30 to 120 connections and 0.40 for the ten largest hubs. The four high-closeness

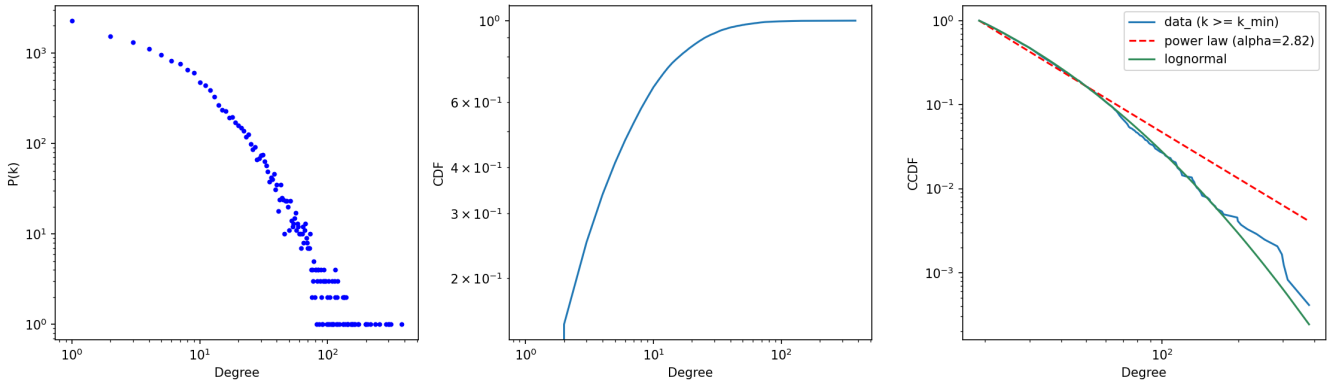


Figure 1: Degree distribution: $P(k)$ on a log-log scale (left), cumulative distribution (center), and complementary cumulative distribution of the tail ($k \geq 19$) with the fitted power law and lognormal (right).

newsletters above range from 0.69 to 0.76, with neighbors in eight or nine of the ten main communities: they are close to the rest of the network because they are connected to all of its parts.

The same measure separates two kinds of newsletters with high betweenness:

- **Bridges between communities.** The newsletters of Junot Díaz, Seymour Hersh and Adam Grant have a moderate degree but a high participation (0.57 to 0.77): their neighbors are spread across communities, so the shortest paths that cross them connect different parts of the network.
- **Brokers within a community.** *Why Is This Interesting?* also ranks high by betweenness (50th), but 81% of its neighbors belong to its own community ($P = 0.34$): it lies on many shortest paths, but mostly within that community.

High betweenness therefore marks a bridge between communities only when it comes with high participation.

Comparison with ER

The properties described above were compared with those of two reference models with the same number of nodes, each generated ten times, in order to identify which of them go beyond chance (Table 3).

The first model is an Erdős–Rényi (ER) random graph, generated with a fixed number of edges so that it matches the real network. In this model every pair of nodes is equally likely to be connected, with a corresponding probability $p = 2L/n(n-1) = 7.1 \times 10^{-4}$, where L is the number of edges, which lies above both the threshold for the emergence of a giant component, $p_c = 1/n = 6.6 \times 10^{-5}$, and the connectivity threshold, $\ln n/n = 6.3 \times 10^{-4}$. The ER graph is therefore in the *connected regime*, although close to its boundary, with 1.6 components on average. At this density an almost connected graph is thus expected by chance: the connectedness of the real network is not a distinctive feature, and is in any case favored by the sampling.

The same holds for distances: the average shortest path of the ER graph (4.32) is almost identical to the real one (4.34), although the real network has a larger diameter (9 against 7). The two differ, instead, in how connections are distributed. In the ER graph degrees follow a Poisson distribution concentrated around the mean, with a maximum of 27 against 380 in the real network, so there are no hubs (Figure 2) and no correlation between the degrees of connected nodes (assortativity 0.00 against −0.08). Its clustering is also about 200 times lower, since two neighbors of a node are no more likely to be connected than any other pair. Randomness alone therefore accounts for the connectivity and the short paths of the real network, but neither for its hubs nor for its clustering.

Comparison with BA

The second model is a Barabási–Albert (BA) graph, in which new nodes attach preferentially to well-connected ones. Since its parameter m must be an integer, $m = 5$ was used, as it gives the number of edges closest to the real network; $m = 6$ leads to the same conclusions (clustering 0.0057, average shortest path 3.60).

Preferential attachment reproduces the heavy tail that the ER model lacks, but not its shape. It overestimates the hubs (maximum degree 470 against 380), consistent with a real tail that bends below a pure power law, and it has no periphery: every node enters with m links, so the minimum degree is 5, whereas 15% of the newsletters have a single connection (Figure 2). The BA graph is also more compact than the real network, with shorter paths (3.78 against 4.34) and a smaller diameter (6 against 9), and it is less disassortative (−0.02

Table 2: Top 10 newsletters by degree, betweenness and closeness.

Degree		Betweenness		Closeness	
heathercoxrichardson	380	heathercoxrichardson	0.044	heathercoxrichardson	0.333
thevccorner	313	vitaliy	0.030	auraist	0.325
22ndcenturyfrontier	302	thevccorner	0.029	paulkrugman	0.316
the-founders-corner	297	auraist	0.027	vitaliy	0.315
themarketdispatch	285	ai-supremacy	0.027	thevccorner	0.314
ai-supremacy	254	22ndcenturyfrontier	0.023	ai-supremacy	0.314
vitaliy	236	the-founders-corner	0.021	thefp	0.311
oldster	217	neweconomies	0.016	honest-broker	0.310
neweconomies	205	doomberg	0.015	the-founders-corner	0.309
macromornings	199	thefp	0.015	noahpinion	0.308

Table 3: Comparison between the real network and the theoretical models (ER, BA). Model values are averages over 10 realizations.

Metric	Real network	ER	BA
Nodes	15,186	15,186	15,186
Edges	82,062	82,062	75,905
Average degree	10.8	10.8	10.0
Maximum degree	380	27	470
Connected components	1	1.6	1
Average clustering	0.1509	0.0007	0.0051
Avg. shortest path	4.34	4.32	3.78
Diameter	9	7	6
Degree assortativity	−0.08	0.00	−0.02

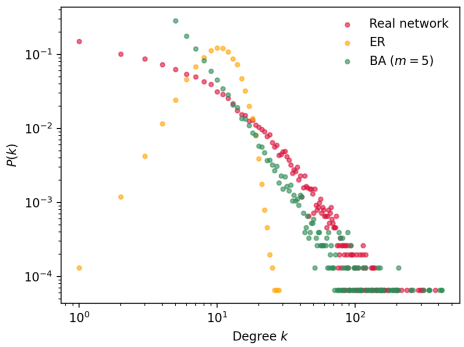


Figure 2: Degree distribution $P(k)$ of the real network compared to the Erdős–Rényi (ER) and Barabási–Albert (BA, $m = 5$) theoretical models on logarithmic scales.

against −0.08). The main difference, however, concerns clustering, which is still about thirty times lower than in the real network.

Taken together, the two comparisons show that the network is a small-world network: it combines the short paths of a random graph with a clustering about two hundred times higher than in the random graph. A preferential-attachment model reproduces the hubs, but not this local structure. Recommendations do not merely target popular newsletters: while major hubs rarely reciprocate, 41% of all recommendations are mutual, suggesting frequent exchange between peers. Section 4 examines whether the local structure of the network reflects topics.

4 OPEN QUESTION

Approach

Having characterized the structure of the network, the analysis turns to its central question: *do newsletters recommend what they write about, or also whom they agree with?* The clustering observed in Section 3 suggests that the network is organized in dense groups. The question is whether these groups simply correspond to topics, or whether recommendations also reflect affinities that topic labels miss, such as a shared political stance.

The same newsletters are grouped twice, in two independent ways:

- **by recommendations**: communities detected from the network alone, ignoring content;
- **by content**: topics inferred from texts alone, ignoring recommendations.

If newsletters recommended only what they write about, the two groupings would largely coincide; where they diverge, the network reveals something that texts alone do not. The comparison is carried out both against a null model that randomly permutes topic labels among nodes and through the inspection of individual communities. Newsletters without a topic label are excluded from the comparison, and their distribution is analyzed separately.

Communities and Their Stability

Communities were detected with the Louvain algorithm, which maximizes modularity. Since Louvain is stochastic, it was run 30 times with different seeds, and the partition most similar on average to the others, measured by normalized mutual information (NMI), was kept as reference. It contains 21 communities, with a modularity of 0.641. The ten largest cover 97.5% of the newsletters (Figure 3); the other eleven are marginal, with 384 newsletters in total.

To assess how much the result depends on the algorithm, the reference partition was compared with 30 further runs of Louvain and 30 runs of Leiden, which guarantees well-connected communities. Agreement is substantial but not complete (NMI 0.73 with Louvain and 0.70 with Leiden): the overall structure is robust, while the boundaries between some communities vary. Table 4 reports, for each community, its *cohesion*, that is, the probability that two of its members end up in the same community in another run.

The most stable communities highlight the network's sharp boundaries. While finance and entrepreneurship might intuitively overlap, they form distinct, highly reproducible groups: *investing* (cohesion 0.90) contains 90% of finance newsletters, whereas *start-ups and technology* (0.83) gathers over 60% of business and technology ones. Both keep roughly three quarters of their edges internal (79% and 75%). Conversely, the *health and science* (0.53), *heterodox-centrist* (0.55) and *literary* (0.56) communities are the least stable among the larger ones, with members frequently split across different communities between runs.

Communities and Topics

The overlap between communities and topics was measured on the 11,886 newsletters with a topic label, using three measures:

- *topical assortativity*, on the macro-topics: whether recommendations connect newsletters on the same topic more often than random links would;
- *normalized mutual information* (NMI): how much the community of a newsletter reveals about its topic;
- *homogeneity* and *completeness*: whether each community contains a single topic, and whether each topic falls within a single community.

To check that this overlap is not a matter of chance, topic labels were shuffled at random among the newsletters, leaving the network and the communities unchanged, 500 times for NMI and 50 for assortativity.

Communities and topics match far more closely than chance would allow. Assortativity is 0.49 and NMI 0.36, whereas the shuffled labels give values close to zero (-0.001 ± 0.002 and 0.003 ± 0.0003). Communities found from the links alone therefore reveal a great deal about what newsletters write about.

Yet topic does not explain everything. Homogeneity (0.38) and completeness (0.35) are both far from 1, which means that the match fails in two ways:

- **Communities mix topics** (low homogeneity). Even in the purest community, music, film and pop culture, the dominant topic covers only 72% of the labeled members; in the heterodox-centrist community it covers 41% (purity in Table 4).
- **Topics are spread over several communities** (low completeness). Finance and food newsletters each fall almost entirely within a single community (90% and 91%), whereas politics is split across three (45%, 24% and 18%), and science and health is the most scattered, with only 41% in its main community.

Where Topic Is Not Enough

Politics is the clearest case of a topic split across communities. Most political newsletters fall into three communities:

- **Liberal mainstream** (*Letters from an American*, Robert Reich, *The Bulwark*): commentary critical of Donald Trump, mostly from the center-left but also from former Republicans.
- **Heterodox-centrist** (*The Free Press*, Andrew Sullivan, *Slow Boring*, *Noahpinion*): writers who reject partisan orthodoxies on either side, especially progressive positions on cultural issues.
- **Anti-establishment** (Chris Hedges, *Racket News*, Caitlin Johnstone): critics of mainstream institutions, media and US foreign policy, drawn from both the left and the right.

The liberal and anti-establishment blocks are among the more stable communities (cohesion 0.69 and 0.72), while the heterodox-centrist one is the least stable of the three political communities (0.55). The anti-establishment community, although the smallest, is also the most inward-looking of the three, keeping 61% of its edges internal.

The split cannot be explained by finer political topics, since the text-based political clusters cut across the three communities:

- **US politics**: most newsletters fall in the liberal community (63%), but 14% belong to the anti-establishment one and 13% to the heterodox one.
- **Geopolitics**: newsletters are divided between the liberal (47%) and the anti-establishment community (30%).
- **UK politics**: the only partial alignment, with half of the cluster in the heterodox community. Its most connected members, however, mix British commentators

Table 4: The ten main communities by size, showing purity (share of the dominant macro-topic) and cohesion (pairwise co-assignment stability under Louvain and Leiden). Newsletters are identified by their URL.

Community	Nodes	Dominant macro-topic	Purity	Cohesion	Central newsletters
Start-ups & technology	2,701	technology	49%	0.83	the-founders-corner, thevccorner
Literary & personal essay	2,667	lifestyle & culture	50%	0.56	oldster, georgesaunders
Investing	2,577	finance	70%	0.90	themarketdispatch, macromornings
Food, fashion & home	1,730	lifestyle & culture	56%	0.64	thesecretingredient, davidlebovitz
Liberal mainstream	1,545	politics	66%	0.69	heathercoxrichardson, robertreich
Heterodox-centrist	1,398	politics	41%	0.55	thefp, andrewsullivan
Anti-establishment	738	politics	56%	0.72	merylnass, racket.news
Health & science	723	science & health	48%	0.53	yourlocalepidemiologist, insidemedicine
Music, film & pop culture	611	arts & letters	72%	0.66	bradkyle, thekevinalexander
Heterogeneous	112	lifestyle & culture	52%	0.39	coolstoriesabout.art, leaderunlock

with American policy writers, suggesting that the cluster captures a register of policy commentary as much as a country.

Health shows a similar pattern. *Science and health* is the most scattered macro-topic, and its two largest shares fall in two different communities:

- **Health and science** (41% of health newsletters): built around public-health and medical newsletters such as *Your Local Epidemiologist* and *Inside Medicine*.
- **Anti-establishment** (11% of health newsletters): of the fifteen most connected among them, 13 publish content critical of COVID-19 vaccines, vaccine mandates or the public-health institutions that promoted them, such as Steve Kirsch's newsletter and *Popular Rationalism*; the other two deal with nutrition and general wellness.

These vaccine-critical newsletters share a topic label with the public-health newsletters, yet they are recommended alongside anti-establishment political commentary rather than alongside other health newsletters. In Figure 3, the two communities lie far apart: the anti-establishment community on the upper edge of the network, the health and science community near its center. In both domains, then, the topic assigned from text is correct but insufficient: the network separates newsletters that write about the same subject from different positions.

Missing Labels

Of the 15,186 newsletters, 3,300 (21.7%) have no topic label: 1,403 have only generic titles, 1,370 have no retrievable text, 387 are not in English and 140 publish mostly video. These newsletters are left without a topic in all the analyses above. The question is therefore: *are newsletters without a topic scattered across the network, or concentrated in some parts of it?* If they are scattered, the results above hold for the whole network; if they are concentrated, the parts where they gather are described less reliably.

To answer this question, the distribution of missing labels was analyzed at two levels:

- **Degree.** As shown in Figure 4, missing labels are concentrated among weakly connected newsletters: the share without a label falls from 35.9% among newsletters with a single connection to 13.0% among those with more than 50. Results show that unlabeled newsletters have fewer connections than labeled ones (median degree 4 against 7), and a Mann–Whitney test confirms that this difference is not due to chance ($p < 10^{-88}$). Among newsletters with a single connection, more than half of the missing labels are due to the absence of text rather than to uninformative titles, which suggests inactive publications or failed downloads.
- **Communities.** The 384 newsletters outside the ten main communities lack a label in 68.5% of cases. Within the main communities, the share of missing labels also varies: it ranges from 12.4% in the food, fashion and home community to 41.1% in the heterogeneous one, and lies between 25.8% and 29.3% in the three political communities. A χ^2 test confirms that these differences are not due to chance ($p < 10^{-50}$), although they are modest in size (Cramér's $V = 0.14$).

Missing labels therefore mainly affect the periphery of the network and, to a lesser extent, some communities (e.g. the political ones).

Side analysis: can communities suggest missing topics? It was checked whether community membership indicates the likely topic of an unlabeled newsletter. On the labeled newsletters of the ten main communities, hiding each label and comparing it with the dominant topic of its community gives 55.8% agreement, against 15.4% by chance and 24.8% when always choosing the most frequent topic. Agreement rises to 68.8% in the three purest communities (investing, liberal mainstream,

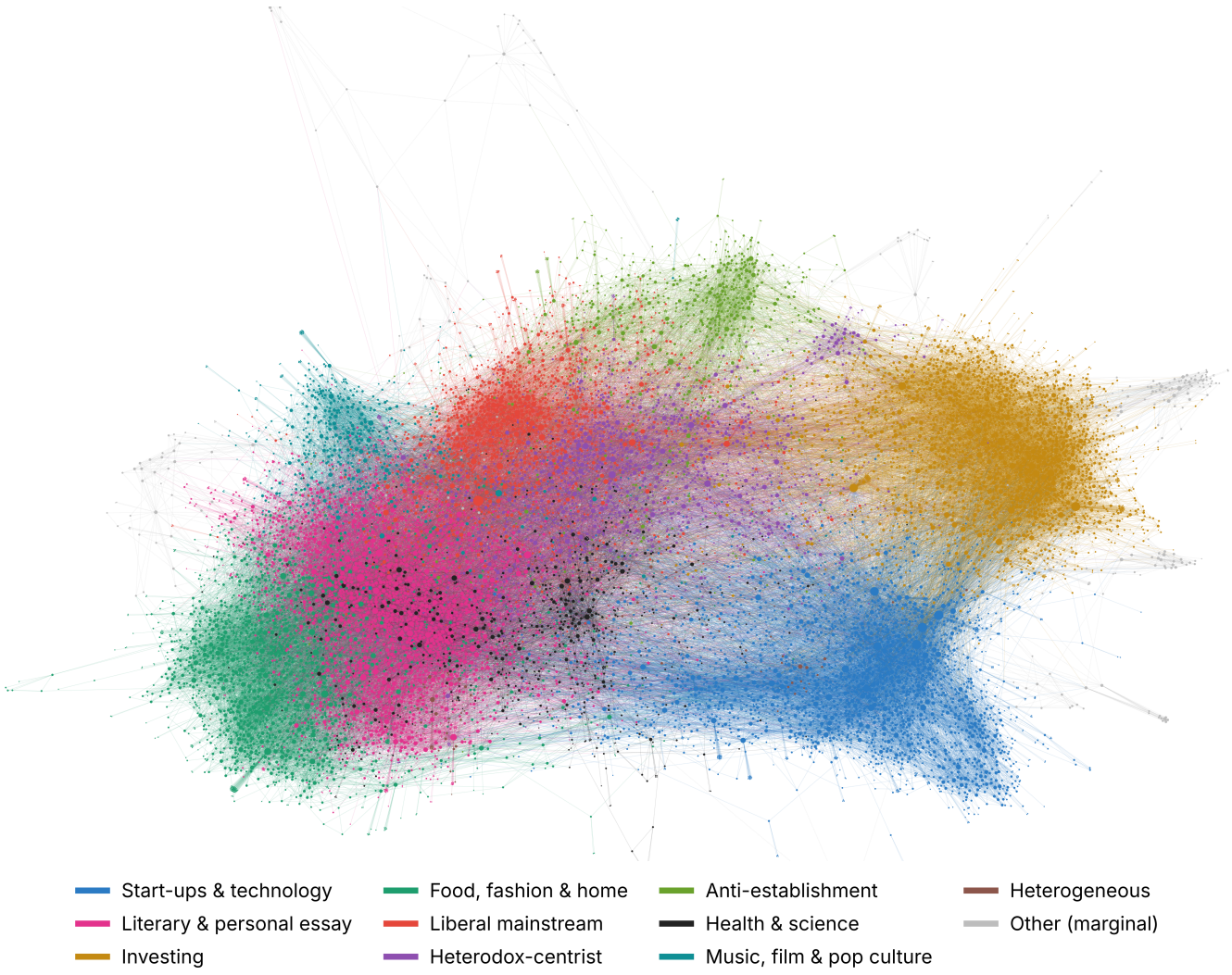


Figure 3: The recommendation network, with newsletters colored by community.

music, film and pop culture), but remains weak in mixed ones such as the heterodox-centrist, anti-establishment and heterogeneous communities.

5 DISCUSSION

This study asked whether Substack recommendations reflect shared subject matter or also capture affinities that topic labels miss. The answer is that they do both, depending on the domain. Topic is the dominant organizing principle: recommendations stay within the same topic far more often than chance would predict, and in domains such as finance and food a single community gathers almost all the newsletters on that subject. Where positions conflict, however, topic is no longer enough. Political newsletters split into three communities that cut across their finer topics, and health newsletters critical of COVID-19 vaccines are recommended alongside anti-establishment political commentary rather than alongside public-health newsletters.

These results extend the observation of Adamic and Glance [1], who found that political blogs link mostly within their own orientation, in three directions. First, the division emerges without any prior labelling: communities were detected from recommendations alone, and stance appeared only on inspection. Second, it is not limited to two opposing camps, since heterodox-centrist newsletters form a community of their own. Third, it appears outside politics: in health, the network separates newsletters on the same subject according to their position.

In the terms of McPherson et al. [4], recommendations combine homophily based on what newsletters write about

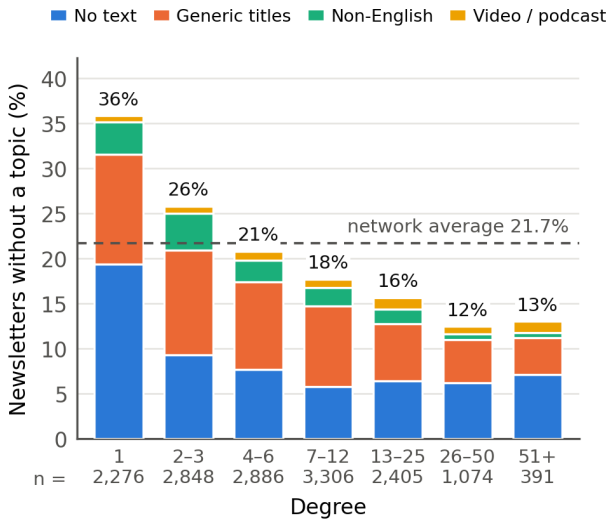


Figure 4: Unlabeled newsletters by degree band and cause of missing label.

with homophily based on the values they express. The anti-establishment community is a clear example: its most central newsletters include left-wing critics of US foreign policy and media as well as vaccine-critical health newsletters, which suggests that what holds it together is a shared distrust of mainstream institutions rather than a common subject or a position on the left–right axis.

The structural analysis is consistent with this interpretation. The network is a small world whose hubs are built by recommendations received, and its clustering is about 200 times higher than in a random graph and 30 times higher than under preferential attachment. Popularity alone, which preferential attachment captures, does not explain the local structure; the community analysis shows that this structure largely follows topic, and in contested domains follows stance.

Limitations

The study has several limitations.

- **Sampling.** The snowball crawl reaches only newsletters connected to the seeds and favors well-recommended ones. Newsletters discovered late have fewer known connections, so centrality partly reflects the order of the crawl.
- **Snapshot and direction.** Since the network is a single snapshot analyzed as undirected, it cannot show how recommendations change over time, nor whether they flow mostly in one direction between communities.
- **Topics from titles.** Topics were inferred from titles, subtitles and the beginning of a few posts, with clusters that are only weakly separated. Some labels are vague (*personal*) or imprecise (*UK politics*), and newsletters that publish mostly video or have generic titles cannot be classified.
- **Missing labels.** Labels are missing mainly in the periphery of the network, so the comparison between communities and topics describes its core more reliably than its margins.
- **Stance is not measured.** Despite the manual analysis carried out, the political and health positions of communities were characterized only qualitatively, from their most central newsletters, without a systematic measure of stance across all members.

Future Work

A directed analysis could show whether recommendations between the political communities are reciprocated or flow mostly one way. Stance could be measured directly from post texts rather than inferred from the core of each community, turning the qualitative characterization used here into a quantitative test. Finally, repeated crawls would show whether these communities are stable over time or drift as the platform grows.

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